INPUT

message_analyzer.cpp > main()

```cpp
#include <iostream>
using namespace std;
#include <cctype>
#include <cstring>
#include <string>

int main () {

    int g = 0, w = 0, y = 0, n = 0, e = 0, t = 0, h = 0, r = 0;

    char ch[50], shout[50], whisp[50];

    cout << " Enter a message (max 50 characters): ";
    cin.getline(ch, 50);


    for (int i = 0; ch[i] != '\0'; i++) {

        g++;
        if (isalpha(ch[i])) {
            if (isupper(ch[i])) {
                w++;

            }else if (islower(ch[i])) {
                y++;
            }

        }else if (isdigit(ch[i])) {
            n++;

        }else if (isspace(ch[i])) {
            e++;

        }else if (ispunct(ch[i])) {
            t++;
        }

    shout[i] = toupper(ch[i]);
    whisp[i] = tolower(ch[i]);

    }

    for (int r = 0; ch[r] != '\0'; r++) {

        if (ch[r] != ' ' && (r == 0 || ch[r-1] == ' ')) {
            h++;
    }
}
```

IMP

```cpp
    shout[g] = '\0';
    whisp[g] = '\0';

    cout << "--- Message Report ---" << endl;
    cout << "Length (manual count) : " << g << endl;
    cout << "Uppercase Letters     : " << w << endl;
    cout << "Lowercase Letters     : " << y << endl;
    cout << "Digits                : " << n << endl;
    cout << "Spaces                : " << e << endl;
    cout << "Special Characters    : " << t << endl;
    cout << "Word Count            : " << h << endl;
    cout << "SHOUTED  : " << shout << endl;
    cout << "whispered : " << whisp << endl;



    return 0;
}
```

OUTPUT

Problems Output Debug Console Terminal Ports

```
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\message_analyzer.exe
 Enter a message (max 50 characters): Hello! I am Raj, 18y/o.
--- Message Report ---
Length (manual count) : 23
Uppercase Letters     : 3
Lowercase Letters     : 10
Digits                : 2
Spaces                : 4
Special Characters    : 4
Word Count            : 5
SHOUTED  : HELLO! I AM RAJ, 18Y/O.
whispered : hello! i am raj, 18y/o.
PS C:\CPE11S4-DOMINGO>
```

EXPLANATION

Problems Output Debug Console Terminal Ports

```
PS C:\CPE11S4-DOMINGO> & C:\CPE11S4-DOMINGO\dumpp.exe
This program takes a message from the user and analyzes the characters that are entered. It uses a loop to check each character and counts the length, uppercase letters, lowercase letters, digits, spaces, and special characters using different conditions. It also has another loop that checks the beginning of each word to get the total word count. The `toupper()` and `tolower()` functions are used to create an uppercase and lowercase version of the original message. After the loops, a null character is added to the end of the new arrays so they can be properly displayed as strings. Finally, the program prints all the counted information together with the SHOUTED and whispered versions of the message.
PS C:\CPE11S4-DOMINGO>
```